

low_tpm

shaikh mohammad kounen

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Low tpm genes across cat

load data

```
cat_species <- read_csv("C:/Users/Lenovo/Documents/mered data/cat/cat.csv",  
  show_col_types = FALSE  
  )
```

- now we have to filter data from merge file into gene who have tpm<1.

```
cat_low_tpm <- cat_species %>%  
  filter(TPM < 1)
```

- now we get no of total low tpm gene.

```
cat("Total number of rows with TPM < 1:", nrow(cat_low_tpm))
```

```
## Total number of rows with TPM < 1: 829450
```

Tissue with most no of Low tpm genes

```
low_tissue <- cat_low_tpm %>%  
  group_by(`Anatomical entity name`) %>%  
  summarise(  
    Low_TPM_Genes = n_distinct(`Gene ID`),  
    .groups = "drop"  
  ) %>%  
  arrange(desc(Low_TPM_Genes))  
print(low_tissue)
```

```
## # A tibble: 17 x 2  
##   'Anatomical entity name'   Low_TPM_Genes  
##   <chr>                     <int>  
## 1 liver                     23968  
## 2 skeletal muscle tissue   23482  
## 3 heart                     22948
```

## 4 adult mammalian kidney	22482
## 5 spleen	22457
## 6 lung	20111
## 7 adipose tissue	19483
## 8 hindlimb muscle	19386
## 9 brain	18116
## 10 testis	14045
## 11 embryo	8755
## 12 tip of external ear	7622
## 13 prefrontal cortex	7181
## 14 zone of skin	7106
## 15 eyeball of camera-type eye	6615
## 16 uterus	6197
## 17 embryonic head	5669

from above table we can conclude that liver has highest no of low tpm genes and embryonic head have lowest

bar-plot on no of low tpm gene across tissue

```
ggplot(low_tissue,
       aes(x = reorder(`Anatomical entity name`, Low_TPM_Genes),
           y = Low_TPM_Genes,
           fill = Low_TPM_Genes)) +

  geom_col(width = 0.8) +

  coord_flip() +

  geom_text(aes(label = Low_TPM_Genes),
            hjust = 1.2,
            color = "white",
            size = 3.8) +

  scale_fill_gradient(
    low = "#90CAF9",
    high = "#0D47A1"
  ) +

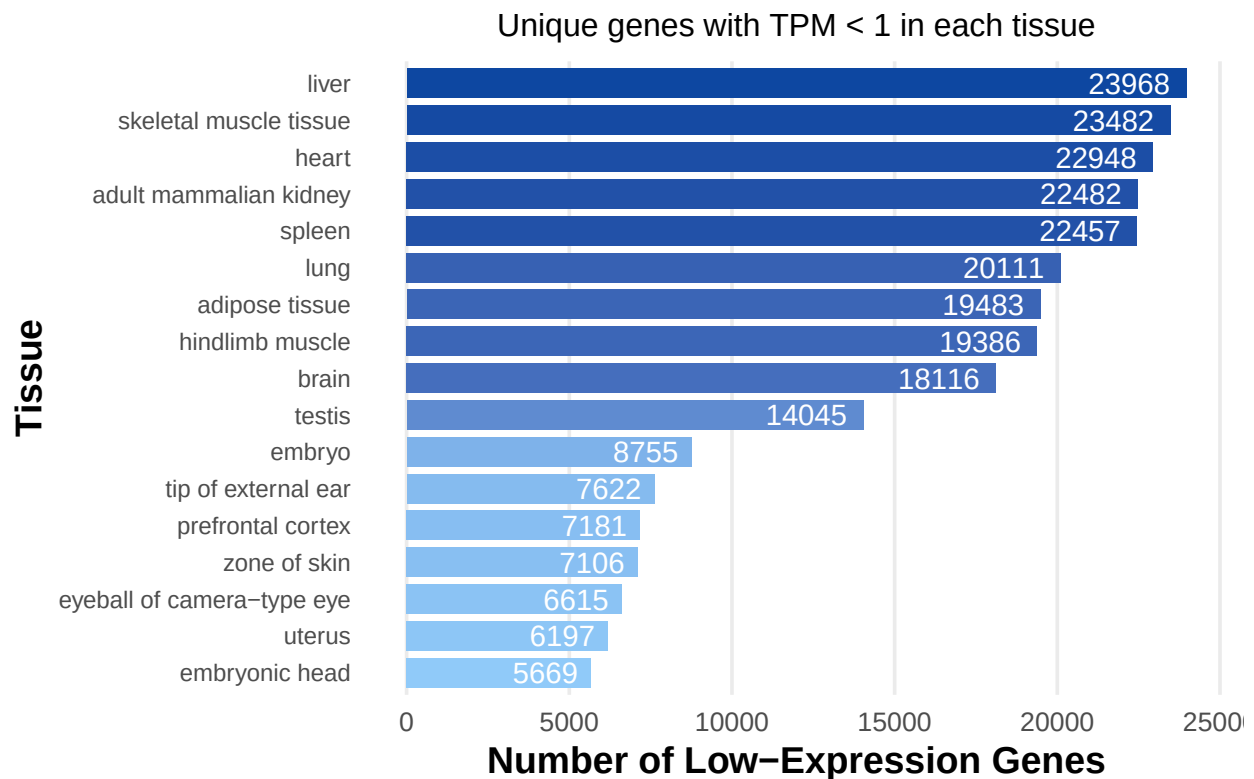
  labs(
    title = "Number of Low-Expression Genes Across Cat Tissues",
    subtitle = "Unique genes with TPM < 1 in each tissue",
    x = "Tissue",
    y = "Number of Low-Expression Genes"
  ) +

  theme_minimal(base_size = 14) +

  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    plot.subtitle = element_text(size = 12, hjust = 0.5),
    axis.title = element_text(face = "bold"),
```

```
axis.text.y = element_text(size = 9),
axis.text.x = element_text(size = 10),
panel.grid.major.y = element_blank(),
panel.grid.minor = element_blank(),
legend.position = "none"
)
```

Number of Low-Expression Genes Across Cat Tissues



conclusion

Low-expression gene analysis (TPM < 1) revealed clear differences among cat tissues. The liver exhibited the highest number of low-expression genes (23,968), whereas the embryonic head had the fewest (5,669). These results demonstrate that the abundance of lowly expressed genes varies across tissues